

Brian Wonjun Lee

Philadelphia, PA | 215-294-7152 | leebwj@sas.upenn.edu | leebrian.dev | [LinkedIn](#)

EDUCATION

University of Pennsylvania

MSE in Computer Graphics & Game Technology (CGGT), Submatriculation

BA in Computer Science & Design (Double Major), Benjamin Franklin Scholars

Relevant Coursework: Interactive Computer Graphics, Computer Animation, 3-D Computer Modeling, Introduction to Algorithms, Introduction to Computer Systems, Programming Languages & Techniques

Philadelphia, PA

Expected May 2028

Expected May 2027

EXPERIENCE

Software Engineer Intern, Aleph Lab (Y Combinator)

June 2026 – Aug 2026 · San Francisco (Remote)

- **Put a full app redesign in front of every user with no rollback — 13 flag-gated PRs** that first restructured the app's information architecture (2x2 home grid, a new Speaking tab, a 5-tab bar), then re-skinned every screen on a **33-section** design system, cleared by a **3,748-test** gate and byte-identical with the flag off
- **Sequenced it so nobody had to relearn the app** — structure shipped incrementally so the layout was familiar before the brand flipped, then a staged **25→50→100%** rollout held each rung across a real class cycle, with experience continuity on so a partial rollout could not change a child's design mid-session
- **Turned a channel that went dark whenever nobody approved a send into one that runs itself** — proposing, clearing its own guardrails and reaching **91% of targeted devices** with **no human in the loop**
- **Produced the first revenue the win-back ever generated**, and reached **~2.7x more users** after proving the eligibility logic had excluded most of its intended audience
- **Made a children's app safe from notification injection** — anyone with a user ID could push arbitrary content until I bearer-authed four public endpoints, **zero caller changes**
- **Stopped the app reporting failures as facts** — a failed request had been telling parents their child spoke in no classes; split error from empty across the growth screens and replaced the blank white crash screen with a branded recovery in Korean and English

Software Engineer Intern, BitMango (Puzzle1 Studio)

June 2025 – Aug 2025 · Pangyo, South Korea

- **Kept 50+ live Unity mobile titles shipping** across a summer of releases, holding UI/UX and gameplay stable
- **Cleared 100+ QA-reported bugs into production**, in Unity prefabs and C# across a portfolio no single engineer owned
- **Kept the portfolio compliant and in the stores** through a platform requirement change, upgrading SDKs fleet-wide

Project Lead & Product Designer, Penn Spark

Jan 2025 – Present · Philadelphia, PA

- **Shipped client-facing 0-to-1 products** leading a cross-functional team of **15+** designers and developers

Database Engineer Intern, IT-Farm

June 2022 – Aug 2022 · Seongnam, South Korea

- **Made semiconductor data from 20+ client companies queryable** — including Samsung, SK and LG — by designing a SQL database prototype managing **1M+ entities** in Oracle and SQLite
- **Cut retrieval time across large-scale datasets** by restructuring and optimizing the queries the team relied on

PROJECTS

Capsule | React, Three.js (R3F), Blender

Jan 2025 – Apr 2025

- **Made memories something you walk through rather than scroll** — the **3D gallery** in React Three Fiber for a collaborative time-capsule app (team of 4), with direct-manipulation placement
- **Owned how the product felt to use** across the create → contribute → unlock flow

Mini Maya | C++, OpenGL, Qt

Feb 2026 – Mar 2026

- **Made arbitrary meshes editable in place** — a **half-edge structure** giving topology-aware traversal, plus an OBJ parser accepting arbitrary n-gons, rendered via OpenGL VBOs
- **Turned coarse geometry into smooth surfaces** with **Catmull–Clark subdivision** and topology ops (split, extrusion, triangulation) in a Qt editor

Mini Minecraft | C++, OpenGL, GLSL

Apr 2026

- **Made a from-scratch voxel world read as varied rather than noisy** — **7-biome procedural generation** (Perlin/FBM/Voronoi) and **3D Perlin caves** in a C++/OpenGL engine (team of 3)
- **Made that world legible** — the GLSL pipeline I owned: PCF shadows, screen-space reflections, vertex AO, day-night cycle

TECHNICAL SKILLS

Languages: TypeScript, JavaScript, Python, Java, C++, C#, SQL, HTML/CSS, OCaml

Frameworks & Libraries: React, Next.js, Node.js, Astro, Tailwind CSS, Three.js, React Three Fiber, JUnit

Developer Tools: Git, GitLab, AWS, MongoDB, SQLite, Oracle, Figma, VS Code, IntelliJ

Graphics & 3D: OpenGL, GLSL, Unreal Engine 5, Unity, Maya, Blender